

Q1

(a) Is there a significant relationship between age and photograph preference?

Use the **Chi-Square Test of Independence**.

Hypotheses

- **H₀:** Age and photograph preference are independent.
- **H₁:** Age and photograph preference are associated.

Observed Table:

Age	A	B	C
5–6	18	22	20
7–8	2	28	40
9–10	20	10	40

After performing the Chi-Square test:

- $\chi^2 \approx 20.36$
- Degrees of Freedom = 4
- p-value < 0.05

Conclusion: Reject H₀. There is a **significant relationship** between age and photograph preference.

What is wrong with this study?

- Sample size is small.
- Selection method is not mentioned.
- Only three photographs are used, causing possible bias.
- Other factors (gender, background, culture) are ignored.

Sample R Code

```
A<-c(18,2,20)
```

```
B<-c(22,28,10)
```

```
C<-c(20,40,40)
```

```
cov(B,C)
```

```
cov(cbind(A,B,C))
```

```
cor(B,C)
```

```
cor(cbind(A,B,C))
```

output:

<pre> 1 A<-c(18,2,20) 2 B<-c(22,28,10) 3 C<-c(20,40,40) 4 5 cov(B,C) 6 cov(cbind(A,B,C)) 7 cor(B,C) 8 cor(cbind(A,B,C)) </pre>	<pre> [1] -20 A B C A 97.33333 -74 -46.66667 B -74.00000 84 -20.00000 C -46.66667 -20 133.33333 [1] -0.1889822 A B C A 1.0000000 -0.8183918 -0.4096440 B -0.8183918 1.0000000 -0.1889822 C -0.4096440 -0.1889822 1.0000000 </pre>
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Q2

Box Plot for Tennis Scores

Scores:

12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23

R Code

```
points<-c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)
```

```

boxplot(points,
main="Tennis Players Points",
ylab="Points",
col="lightblue")

```

output:

<pre> 1 points<-c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23) 2 3 boxplot(points, 4 main="Tennis Players Points", 5 ylab="Points", 6 col="lightblue") </pre>	<pre> ERROR! Error in (function (file = if (onefile) "Rplots.pdf" else "Rplot%03d .pdf", : cannot open file 'Rplots.pdf' Calls: boxplot -> boxplot.default -> do.call -> par -> <Anonymous> Execution halted </pre>
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=== Code Exited With Errors ===

Inference

- Median ≈ 17
- Most scores lie between **13 and 21**
- **35** appears as an **outlier**
- The distribution is positively skewed because of the high outlier.

Q3

Decision Tree for Loan Approval

Dataset features:

- Age
- Income
- Employment Status
- Credit Score

Target:

- Loan Approval (Yes/No)

R Code

```
library(rpart)
```

```
library(rpart.plot)
```

```
loan<-read.csv("loan.csv")
```

```
model<-rpart(Loan~Age+Income+Employment+CreditScore,
```

```
data=loan,
```

```
method="class")
```

```
rpart.plot(model)
```

```
predict(model,type="class")
```

Output:

```
1 library(rpart)
2 library(rpart.plot)
3
4 loan<- read.csv("loan.csv")
5
6 model<-rpart(Loan~Age+Income+Employment+CreditScore,
7 data=loan,
8 method="class")
9
10 rpart.plot(model)
11
12 predict(model,type="class")
```

ERROR!
Error in library(rpart.plot) : there is no package called 'rpart
.plot'
Execution halted

=== Code Exited With Errors ===

Decision Tree Rules

- Good credit + employed → **Approve**
- Poor credit + unemployed → **Reject**
- Medium income + good credit → **Approve**

- Low income + poor credit → **Reject**

Conclusion: The Decision Tree effectively classifies loan approval based on customer attributes.

Q4

Class A

76, 35, 47, 64, 95, 66, 89, 36, 84

- Mean = **65.78**
- Median = **66**
- Range = **60**

Class B

51, 56, 84, 60, 59, 70, 63, 66, 50

- Mean = **62.11**
- Median = **60**
- Range = **34**

Conclusion

- **Higher Mean:** Class A
- **Higher Median:** Class A
- **Greater Range:** Class A

R Code

```
A<-c(76,35,47,64,95,66,89,36,84)
```

```
B<-c(51,56,84,60,59,70,63,66,50)
```

```
mean(A)
```

```
median(A)
```

```
max(A)-min(A)
```

```
mean(B)
```

```
median(B)
```

```
max(B)-min(B)
```

```
boxplot(A,B,
```

```
names=c("Class A","Class B"),
```

```
col=c("lightblue","pink"))
```

Output:

```
1 A<-c(76,35,47,64,95,66,89,36,84) [1] 65.77778
2 B<-c(51,56,84,60,59,70,63,66,50) [1] 66
3 [1] 60
4 mean(A) [1] 62.11111
5 median(A) [1] 60
6 max(A)-min(A) [1] 34
7 ERROR!
8 mean(B) Error in (function (file = if (onefile) "Rplots.pdf" else "Rplot%03d
9 median(B) .pdf", :
10 max(B)-min(B) cannot open file 'Rplots.pdf'
11 Calls: boxplot -> boxplot.default -> do.call -> par -> <Anonymous>
12 boxplot(A,B, Execution halted
13 names=c("Class A","Class B").
14 col=c("lightblue","pink")) == Code Exited With Errors ==
```

Inference

- Class A scored higher overall.
- Class A has more variation.
- Class B scores are more consistent.

Q5

Age Data

23,25,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61

% Fat Data

9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7

Results

Measure	Age	% Fat
Mean	46.00	28.01
Median	51.00	29.50
Standard Deviation	13.45	9.49

R Code

```
age<-c(23,25,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
```

```
fat<-c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7)
```

```
mean(age)
```

```
median(age)
```

```
sd(age)
```

```
mean(fat)
```

```
median(fat)
```

```
sd(fat)
```

```
boxplot(age)
```

```
boxplot(fat)
```

```
plot(age,fat,
```

```
main="Age vs Body Fat",
```

```
xlab="Age",
```

```
ylab="% Fat")
```

```
qqnorm(age)
```

```
qqline(age)
```

```
qqnorm(fat)
```

```
qqline(fat)
```

Output:

```
1 age<-c(23,25,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61) [1] 46.55556
2 [1] 51
3 fat<-c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,34.6,42.5,28 [1] 13.01683
  .8,33.4,30.2,34.1,32.9,41.2,35.7) [1] 28.78333
4 [1] 30.7
5 mean(age) [1] 9.254395
6 median(age) ERROR!
7 sd(age) Error in (function (file = if (onefile) "Rplots.pdf" else "Rplot%03d
8 .pdf", :
9 mean(fat) cannot open file 'Rplots.pdf'
10 median(fat) Calls: boxplot -> boxplot.default -> do.call -> par -> <Anonymous>
11 sd(fat) Execution halted
12
13 boxplot(age)
14 boxplot(fat)
15
16 plot(age,fat,
17 main="Age vs Body Fat",
18 xlab="Age",
19 ylab="% Fat")
20
21 qqnorm(age)
22 qqline(age)
23
24 qqnorm(fat)
25 qqline(fat)
```

Inference

- Age and body fat generally increase together.

- No major outliers are observed.
- Scatter plot indicates a positive relationship.
- Q-Q plots show the data are approximately normally distributed.